

# GOSFORD HIGH SCHOOL



**Year 11**

## MATHEMATICS EXTENSION 1

### Preliminary Course

### Assessment Task 1

**2010**

*Time Allowed: 70 minutes*

#### Instructions:

- There are 5 questions.
- Answer each question on your own paper showing all necessary working.
- Start **EACH** question on a new sheet of paper.
- Calculators may be used.

| Question                | Mark |
|-------------------------|------|
| 1. Arithmetic / Algebra | /10  |
| 2. Polynomials          | /10  |
| 3. Counting Theory      | /10  |
| 4. Inequalities         | /10  |
| 5. Further Algebra      | /10  |
| TOTAL                   | /50  |

**QUESTION 1:** (10 Marks)

- |    |                                                                                                                                                                                                                                                                                                          | Marks |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| a. | Calculate:                                                                                                                                                                                                                                                                                               | 2     |
|    | $\frac{76.5}{17 \times (81.76)^3} \times \frac{1851.7 - 24^2}{\sqrt{47.8}}$                                                                                                                                                                                                                              |       |
|    | writing your answer in scientific notation, correct to 3 significant figures.                                                                                                                                                                                                                            |       |
| b. | A builder intends to pay cash for building materials and receives a discount of $12 \frac{1}{2} \%$ for doing so. As well, he receives a further 10% trade discount on this cash price. If he eventually pays \$1055, find the original price of the building materials. (Answer to the nearest dollar.) | 2     |
| c. | Evaluate $\left( \frac{2^x + 2^x + 2^x + 2^x}{2^{x+1}} \right)^2$                                                                                                                                                                                                                                        | 2     |
| d. | If $\frac{a^2 b^2}{c^4} \times \frac{c^3}{a^5 b^3} \times \frac{(a^2 c)^3}{b^2} = a^x b^y c^z$ ,<br>find the value of $x, y, z$                                                                                                                                                                          | 2     |
| e. | If $(2\sqrt{3} - \sqrt{7})^3 = m\sqrt{3} + n\sqrt{7}$ ,<br>find the value of $m, n$                                                                                                                                                                                                                      | 2     |

**QUESTION 2:** (10 Marks)

|                                                                                                                                                                         | Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| a. For the polynomial $P(x) = 2x^4 - x^2 + x - 3$<br>State the                                                                                                          |       |
| i. Degree                                                                                                                                                               | 1     |
| ii. Constant term                                                                                                                                                       |       |
| b. Which of the expressions:<br><br>A. $5 - x^2 - x^3$ B. $7x^2 + \frac{7}{x^2}$<br><br>C. 3                      D. $x\sqrt{x} + \sqrt{x}$<br><br>are polynomials?     | 1     |
| c. What is a monic polynomial? Explain <b>and</b> give an example.                                                                                                      | 1     |
| d. Divide $P(x) = x^4 + x^3 - x^2 + x - 2$ by $x^2 - 1$ and express your answer in the form<br><br>$P(x) = A(x)Q(x) + R(x)$                                             | 2     |
| e. Find the zeros of the polynomial<br><br>$Q(x) = x^4 - 2x^3 - 8x^2$<br><br>and use them to sketch the graph of $y = Q(x)$<br>(Indicate all intercepts on your sketch) | 2     |
| f. Show that $x - 1$ is <b>not</b> a factor of<br><br>$G(x) = x^5 - 2x^4 + 7x^2 - 3x + 5$                                                                               | 1     |
| g. When $F(x) = ax^2 - 3x + 1$ and $H(x) = x^3 - 3x + 2$ are divided by $x + 1$ they have the same remainder. Find the value of $a$ .                                   | 2     |

### QUESTION 3: (10 Marks)

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|                                                                                                                              | Marks |
|------------------------------------------------------------------------------------------------------------------------------|-------|
| a. Show <b>how</b> you would calculate ${}^6P_4$ . (Do <b>not</b> evaluate)                                                  | 1     |
| b. Two different numbers are chosen at random from 1, 2, 3, 4, 5                                                             | 1     |
| i. How many different <b>combinations</b> can be chosen?                                                                     | 1     |
| ii. How many of these combinations have an even sum when the two numbers are added together?                                 | 1     |
| c. Let each different arrangement of all the letters of INTERESTED be called a word. How many words are possible?            | 1     |
| d. In how many ways can 6 girls sit at a round table if:                                                                     | 1     |
| i. There are no restrictions                                                                                                 | 1     |
| ii. 3 particular girls insist on sitting together                                                                            | 1     |
| e. A ticket queue has 5 men and 5 women standing in a line. Find how many different arrangements of the line are possible if |       |
| i. the men and women can stand anywhere in the line                                                                          | 1     |
| ii. the men and women alternate                                                                                              | 1     |
| iii. 2 particular women cannot stand together.                                                                               | 2     |

**QUESTION 4:** (10 Marks)

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|                                                               | Marks |
|---------------------------------------------------------------|-------|
| a. Solve $\frac{4}{5-x} \leq 1$                               | 2     |
| b. Solve $\frac{x^2-4}{x} > 0$                                | 2     |
| c. Solve $\frac{5x+9}{ x-5 } > 1$                             | 3     |
| d. Solve and graph the solution(s) to:<br>$ x+4  +  x-4  = 8$ | 3     |

**QUESTION 5:** (10 Marks)

|                                                                                                                              | Marks |
|------------------------------------------------------------------------------------------------------------------------------|-------|
| a. Express as a single fraction:<br>$(x^{-1} + y^{-1})^{-1}$                                                                 | 2     |
| b. If $x = \frac{3 - 2\sqrt{2}}{3 + 2\sqrt{2}}$ , express $x - \frac{1}{x}$<br>in simplest form with a rational denominator. | 3     |
| c. Simplify:<br>$\frac{20x + 12y - 25x^2 + 9y^2}{25x^2 - 9y^2} \div \frac{8 - 10x + 6y}{25x^2 - 30xy + 9y^2}$                | 3     |
| d. i. Express $x^4 + 4$ in the form<br>$(x^2 + a)^2 - (bx)^2$<br>and find the value of $a$ and $b$                           | 1     |
| ii. Hence, express $x^4 + 4$ as a product of two quadratic factors                                                           | 1     |

**END OF TEST**

Q1,

a)  $0.001519233... (\text{calc.})$   
 $= 1.52 \times 10^{-3}$

b) Amount paid:  
 $90\% \text{ of } 87\frac{1}{2}\%$   
 $= 78\frac{3}{4}\%$   
 $\therefore 78\frac{3}{4}\% = \$1055$   
 $1\% = \$13.397...$   
 $100\% = \$1339.7...$

$\therefore$  Original Price = \$1340

c)  $\left(\frac{2^x + 2^x + 2^x + 2^x}{2^{x+1}}\right)^2$   
 $= \left(\frac{4 \cdot 2^x}{2^{x+1}}\right)^2$   
 $= \left(\frac{2^2 \cdot 2^x}{2^{x+1}}\right)^2$   
 $= \left(\frac{2^{x+2}}{2^{x+1}}\right)^2$   
 $= 2^2$   
 $= 4$

d)  $\frac{a^2 b^2}{c^4} \times \frac{c^3}{a^5 b^3} \times \frac{(a^3 c)^3}{b^2}$   
 $= \frac{a^2 b^2 c^3}{a^5 b^3 c^4} \times \frac{a^6 c^3}{b^2}$   
 $= \frac{a^8 b^2 c^6}{a^5 b^5 c^4} = \frac{a^3 c^2}{b^3}$

$\therefore x = 3, y = -3, z = 2$

e)  $(2\sqrt{3} - \sqrt{7})^3$   
 $= (2\sqrt{3} - \sqrt{7})(2\sqrt{3} - \sqrt{7})^2$   
 $= (12 - 4\sqrt{21} + 7)(2\sqrt{3} - \sqrt{7})$   
 $= (19 - 4\sqrt{21})(2\sqrt{3} - \sqrt{7})$   
 $= 38\sqrt{3} - 19\sqrt{7} - 8\sqrt{63} + 4\sqrt{147}$   
 $= 38\sqrt{3} - 19\sqrt{7} - 24\sqrt{7} + 28\sqrt{3}$   
 $= 66\sqrt{3} - 43\sqrt{7}$   
 $\therefore m = 66, n = -43$

Q2,

a) (i) Degree = 4  $\left(\frac{1}{2}\right)$   
(ii) Constant = -3  $\left(\frac{1}{2}\right)$

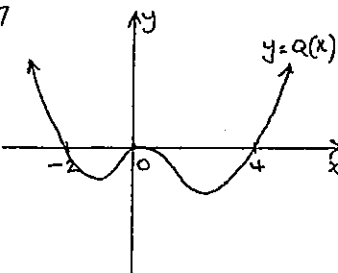
b) A and C (i)  
c) A monic polynomial is a polynomial with the leading co-efficient equal to 1.  $\left(\frac{1}{2}\right)$

eg:  $x^3 - 4x^2 + 5x - 7$   $\left(\frac{1}{2}\right)$   
(of course others)

d)  $\frac{x^2 + x}{x^2 - 1} \div \frac{x^4 + x^3 - x^2 + x - 2}{x^4 - x^2}$   
 $= \frac{x^2 + x}{x^2 - 1} \times \frac{x^4 - x^2}{x^4 + x^3 - x^2 + x - 2}$   
 $= \frac{x^2 + x}{x^2 - 1} \times \frac{x^2(x^2 - 1)}{x^2(x^2 + x^3 - x^2 + x - 2)}$   
 $= \frac{x^2 + x}{x^2 - 1} \times \frac{x^2(x^2 - 1)}{x^2(x^2 + x^3 - x^2 + x - 2)}$   
 $= \frac{x^2 + x}{x^2 - 1} \times \frac{x^2(x^2 - 1)}{x^2(x^2 + x^3 - x^2 + x - 2)}$

$\therefore P(x) = (x^2 - 1)(x^2 + x) + (2x - 2)$

e)  $Q(x) = 0$  for zeros  
i.e.  $x^2(x^2 - 2x - 8) = 0$   
 $x^2(x - 4)(x + 2) = 0$   
 $\therefore$  zeros = 0, 4, -2



f)  $G(1) = 1 - 2 + 7 - 3 + 5 = 8$

Since  $G(1) \neq 0$   
 $x - 1$  is not a factor

g)  $F(-1) = a + 3 + 1 = a + 4$   
 $H(-1) = -1 + 3 + 2 = 4$

If the same remainder;  
 $\therefore a + 4 = 4$   
 $a = 0$

Q3,

a) Using  ${}^n P_r = \frac{n!}{(n-r)!}$   
 ${}^6 P_4 = \frac{6!}{(6-4)!} = \frac{6!}{2!}$   
 $= 6 \times 5 \times 4 \times 3$

Q3, b) (i)  ${}^5 C_2 = 10$   
(ii) either 2 odds or 2 evens  
i.e. (2+4), (1+3), (3+5)  
(1+5)

Answer = 4

c)  $\frac{10!}{2!3!} = 302400$

d) (i) using  $(n-1)!$   
 $= 5!$   
 $= 120$  ways.

(ii) Grouping 3 girls together in a circle  
 $= (4-1)!$   
 $= 3!$  and

the 3 girls can be placed in  $3!$  ways  
total ways =  $3! \times 3!$   
 $= 36$

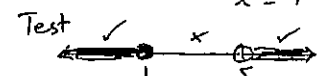
(e) (i)  $10! = 3628800$   
(ii) Men  $\rightarrow 5!$   
Women  $\rightarrow 5!$

Now either  
mwmwmwmwmw  
or  
wmwmwmwmwmw  
 $\therefore$  Ways =  $2 \times 5! \times 5!$   
 $= 28800$   
(iii) 2 particular women together =  $2 \times 9!$

$\therefore$  not together  
 $= 10! - 2 \times 9!$   
 $(3628800 - 806400)$   
 $= 2822400$

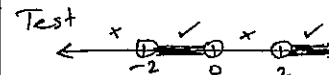
Q4 a) Consider  $\frac{4}{5-x} = 1$

C.V:  $x \neq 5$   
 $4 = 5 - x$   
 $x = 1$



$\therefore x \leq 1, x > 5$

b) Consider  $\frac{x^2 - 4}{x} = 0$   
C.V:  $x \neq 0$   
 $x^2 - 4 = 0$   
 $x = \pm 2$

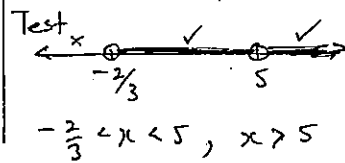


$\therefore -2 < x < 0, x > 2$

c) Consider  $\frac{5x+9}{|x-5|} = 1$   
 $x \neq 5$   
 $5x+9 = |x-5|$

$5x+9 = x-5$  or  $5x+9 = -x+5$   
 $4x = -14$   $6x = -4$   
 $x = -\frac{7}{2}$   $x = -\frac{2}{3}$

check  $x = -\frac{2}{3}$  is only soln to equation



d)  $|x+4| + |x-4| = 8$   
4 cases:  $x+4 > 0$  and  $x-4 > 0$

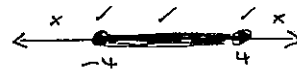
$x+4 + x-4 = 8$   
 $2x = 8$   
 $x = 4$  ✓

2,  $x+4 < 0$  and  $x-4 < 0$   
 $-(x+4) - (x-4) = 8$   
 $-x-4 - x+4 = 8$   
 $-2x = 8$   
 $x = -4$  ✓

3,  $x+4 > 0$  and  $x-4 < 0$   
 $(x+4) - (x-4) = 8$   
 $x+4 - x+4 = 8$   
 $x+4 - x+4 = 8$   
 $8 = 8$

i.e. in this region  $-4 < x < 4$  equation is always true

4,  $x+4 < 0$  and  $x-4 > 0$   
 $-(x+4) + (x-4) = 8$   
 $-x-4 + x-4 = 8$   
 $-8 = 8$   
 $\therefore$  no solutions for  $x < -4$  and  $x > 4$



4 a)  $(x^{-1} + y^{-1})^{-1}$   
 $= \left(\frac{1}{x} + \frac{1}{y}\right)^{-1}$   
 $= \left(\frac{y+x}{xy}\right)^{-1}$   
 $= \frac{xy}{x+y}$

Q.5 b)

$$x - \frac{1}{x} = \frac{3-2\sqrt{2}}{3+2\sqrt{2}} - \frac{3+2\sqrt{2}}{3-2\sqrt{2}}$$

$$= \frac{(3-2\sqrt{2})^2 - (3+2\sqrt{2})^2}{(3+2\sqrt{2})(3-2\sqrt{2})}$$

$$= \frac{9 - 12\sqrt{2} + 8 - (9 + 12\sqrt{2} + 8)}{9 - 8}$$

$$= \frac{17 - 12\sqrt{2} - 17 - 12\sqrt{2}}{1}$$

$$= -24\sqrt{2}$$

$$c) \frac{20x + 12y - 25x^2 + 9y^2}{25x^2 - 9y^2} \times \frac{25x^2 - 30xy + 9y^2}{8 - 10x + 6y}$$

$$= \frac{4(5x+3y) - (5x-3y)(5x+3y)}{(5x-3y)(5x+3y)} + \frac{(5x-3y)(5x-3y)}{2(4-5x+3y)}$$

$$= \frac{(5x+3y)(4-5x+3y)}{(5x-3y)(5x+3y)} \times \frac{(5x-3y)(5x-3y)}{2(4-5x+3y)}$$

$$= \frac{5x-3y}{2}$$

$$d) (i) x^4 + 4 = x^4 + 4x^2 + 4 - 4x^2 \\ = (x^2 + 2)^2 - (2x)^2$$

$$\therefore a = 2, b = 2$$

$$(ii) x^4 + 4 =$$

$$\left[ (x^2 + 2) - 2x \right] \left[ (x^2 + 2) + 2x \right]$$

$$= (x^2 - 2x + 2)(x^2 + 2x + 2)$$